

1. Question

Design a “Farm Animals” Java application with the details of animals like Cow, Pig, and Horse. Consider the following details like where they stay, what they eat, and the sound they make by using classes and objects.

The program should demonstrate the use of classes, objects, constructors, methods, and method overloading.

2. Steps for Execution

1. Install Java JDK on the system.

2. Open Eclipse, IntelliJ IDEA, NetBeans, VS Code, or any Java IDE.

3. Create a Java file named:

FarmAnimals.java

4. Enter the given Java program.

Save the program.

5. Compile the program:

```
javac FarmAnimals.java
```

6. Run the program:

```
java FarmAnimals
```

The program creates a FarmAnimal class.

7. Create objects for:

Cow

Pig

Horse

8. Store the name, place where the animal stays, and food it eats.

9. Display the details of each animal using the display() method.

10. Display the sound of the animals using the sound() method.

11. For the horse, the overloaded sound() method is called with the number of times the sound is made.

13. Display all animal details and sounds on the console.

3. Algorithm

Step 1: Start the program.

Step 2: Create a class named FarmAnimal.

Step 3: Declare the following data members:

name

place

food

Step 4: Create a default constructor and assign default values.

Name = Unknown

Place = Farm

Food = Grass

Step 5: Create a parameterized constructor to initialize the animal's name, place, and food.

Step 6: Create the display() method to display the animal details.

Step 7: Create a sound(String sound) method to display the sound.

Step 8: Create an overloaded sound(String sound, int times) method to display the sound and number of times it is made.

Step 9: Create a Cow object with:

Name = Cow

Place = Cowshed

Food = Grass

Step 10: Create a Pig object with:

Name = Pig

Place = Pigsty

Food = Corn

Step 11: Create a Horse object with:

Name = Horse

Place = Stable

Food = Hay

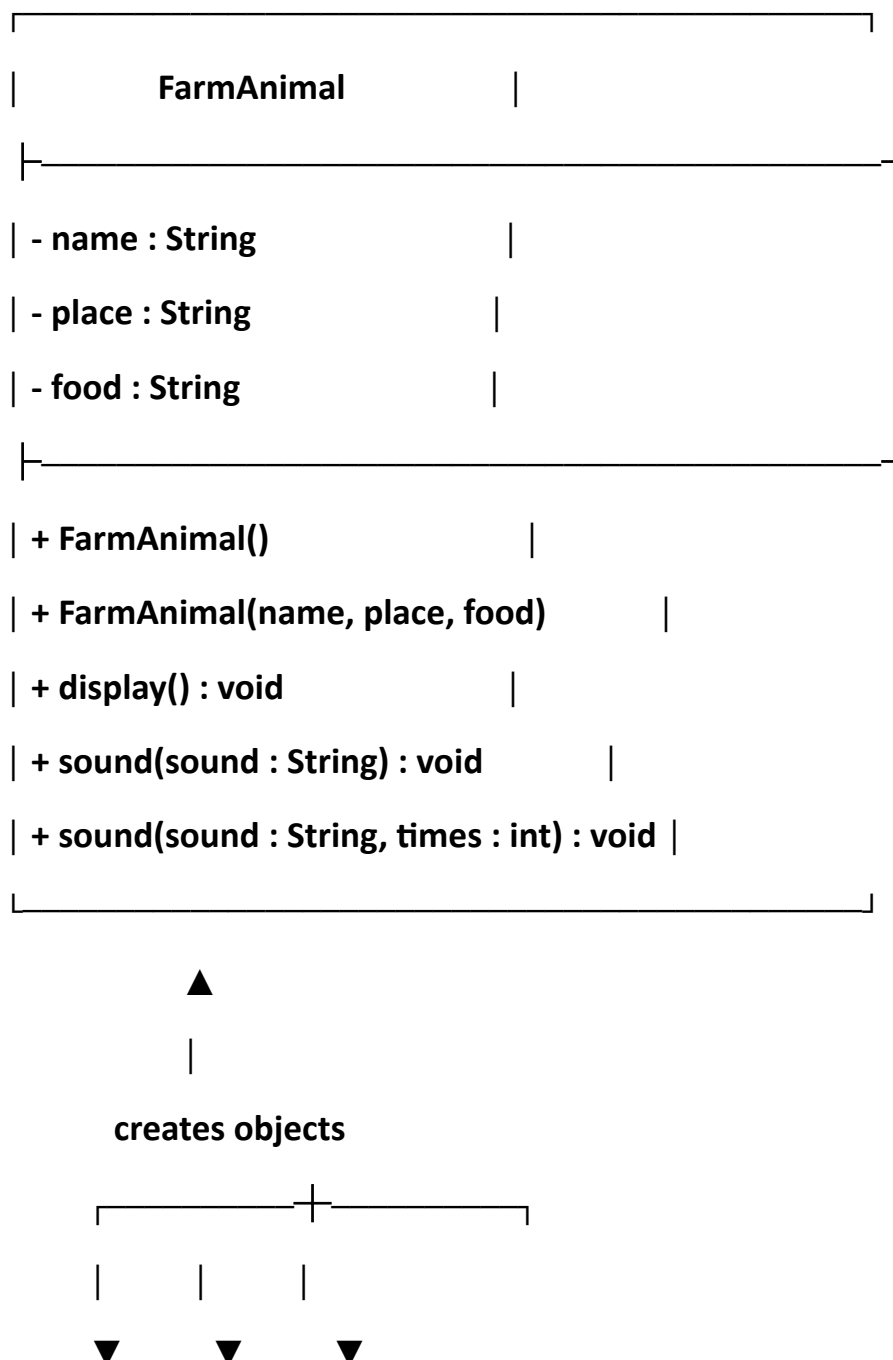
Step 12: Display the Cow details and sound.

Step 13: Display the Pig details and sound.

Step 14: Display the Horse details and sound three times.

Step 15: Stop the program.

UML CLASS DIAGRAM:



cow pig horse

CODE:

```
package javacore;

class FarmAnimal {

    String name;

    String place;

    String food;

    FarmAnimal() {

        name = "Unknown";

        place = "Farm";

        food = "Grass";

    }

    FarmAnimal(String name, String place, String food) {

        this.name = name;

        this.place = place;

        this.food = food;

    }

    void display() {

        System.out.println("Animal: " + name);

        System.out.println("Lives in: " + place);

        System.out.println("Eats: " + food);

    }

    void sound(String sound) {

        System.out.println("Sound: " + sound);

    }

    void sound(String sound, int times) {

        System.out.println("Sound: " + sound);

    }

}
```

```
        System.out.println("Number of times: " + times);
    }
}

public class FarmAnimals {

    public static void main(String[] args) {

        FarmAnimal cow = new FarmAnimal(
            "Cow", "Cowshed", "Grass");

        FarmAnimal pig = new FarmAnimal(
            "Pig", "Pigsty", "Corn");

        FarmAnimal horse = new FarmAnimal(
            "Horse", "Stable", "Hay");

        cow.display();
        cow.sound("Moo");
        System.out.println();
        pig.display();
        pig.sound("Oink");
        System.out.println();
        horse.display();
        horse.sound("Neigh", 3);
    }
}
```

OUTPUT SCREENSHOT (SAMPLE RUN):

Animal: Cow

Lives in: Cowshed

Eats: Grass

Sound: Moo

Animal: Pig

Lives in: Pigsty

Eats: Corn

Sound: Oink

Animal: Horse

Lives in: Stable

Eats: Hay

Sound: Neigh

Number of times: 3